Fiscal Year:	FY 2021	**Task Last Updated:**	FY 08/27/2021
PI Name:	Chahine, Georges D.Sc.		
Project Title:	Two-Phase Flow Separation and Characterization for Microgravity Applications		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	20794	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2008 Fluid Physics NNH08ZTT002N
Start Date:	10/21/2015	**End Date:**	12/31/2020
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/31/2020 per NSSC information (Ed., 9/9/20) NOTE change in grant number to 80NSSC18M0001 per NSSC information and J. McQuillen/GRC (Ed., 1/16/2018)		
Key Personnel Changes/Previous PI:	September 2019 report: Dr's. Gregory Loraine and Jingsen Ma have been added as Co-Investigators.		
COI Name (Institution):	Loraine, Gregory Ph.D. (Dynaflow, Inc.) Hsiao, Chao-Tsung (Dynaflow, Inc.) Ma, Jingsen (Dynaflow, Inc.)		
Grant/Contract No.:	80NSSC18M0001 ; NNX16AB85A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE: Continuation of "Microgravity Fluid Separation: Dynaswirl® Bubble Vortex Separator," grant NNX11AO76A with new grant NNX16AB85A and title, "Two-Phase Flow Separation and Characterization for Microgravity Applications" and same Principal Investigator Georges Chahine, D.Sc. (Ed., 4/18/2016); 1/16/2018 update: New grant number 80NSSC18M0001 awarded for same title--Two-Phase Flow Separation and Characterization for Microgravity Applications (Ed., 1/16/18) The limits on the amount of liquids and gases carried to space make it imperative to recycle and reuse. On Earth, bubbles are relatively easily separable by buoyancy. In microgravity, other external forces such as a centrifugal force must be utilized to separate bubbles from liquids. Two categories of centrifugal force separators exist: one uses active rotation by mechanically spinning the tank. This is very efficient but requires shaft, bearings, and motor. In the second, the tank is fixed and the rotation is induced by eccentric injection of the mixture (Free Vortex Separator or FVS). This FVS has been proven very efficient for mid-range void fraction (50-80% gas) with an efficiency approaching 100%. However, this FVS is not as efficient for lower void fraction, and there is a need of a separator for low void fractions. We have developed a phase separator at DYNAFLOW based on a DYNASWIRL® swirl flow nozzle. It consists of two concentric cylinders, the fluid is introduced in the space between the two cylinders, and the swirling flow inside the inner cylinder is generated with the help of tangential injection slots. With this configuration one can make the vortex circulation intense enough to generate cavitation with a very stable vaporous/gaseous core even at low flow rates. Operation with the high swirl near or slightly below cavitation inception in the vortex core guarantees removal of air even from mixtures with very low void fractions. This is achieved by forcing the bubbles to expand and collect in the vortex core. These are then removed through two central exit orifices at each end of the cylinder. This configuration was tested successfully during a reduced-gravity flight and we are planning additional testing. We are further developing and experimentally testing this configuration to characterize its gas-liquid separation performance and to improve its efficiency. We are also conducting numerical simulations to help optimize the design and ultimately build a compact system for space validation on the International Space Station (ISS) using the existing Fluid Integrated Rack facility.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	As a passive separator device, the DYNASWIRL® phase separator has the advantage of simplicity and dependability. It requires no moving mechanical parts and low power. Compared to other passive separator devices, which usually are not very efficient at low void fraction, the DYNASWIRL® is able to force low void fraction gas out of the liquid into the central core of the vortex by combining the effects of swirl, cavitation, and rectified gas diffusion. In addition to being an effective phase separator for space applications, it will also find wide applications in Earth applications where fast separation of low void fraction gas is of concern.
Task Progress:	The DYNASWIRL® separator uses swirl and cavitation to force gas out of two-phase mixtures into the central core of the generated vortex. The two-phase mixture is forced through tangential slots into a cylindrical chamber to create a strong swirling flow. The injection slots and the exit orifice are designed to generate an intense vortex swirl, which can lead to early cavitation on the vortex axis and enhancement of dissolved gas diffusion into the bubbles. This leads to shortened times to move the bubbles into the vortex core through centripetal force. The resulting gaseous core can then be very stable even at low flow rates. This gas can then be extracted from the core from one end of the chamber or through a tubular gas permeable membrane located at the chamber axis, while the liquid devoid of bubbles is extracted at the other end. Under the present grant, we have continued our efforts to further investigate the performance of the separator and improve the DYNASWIRL® through experimental testing and numerical simulations aiming at optimizing the design and at ultimately developing a fully operational separator for space applications. During the first two years of this project, most efforts were directed towards preparations for the participation of a DYNASWIRL® separator in the NASA planned Two Phase Flow Separator Experiment (TPFSE), which was supposed to be conducted on the International Space Station (ISS). Specifications of the operation parameters were drawn, a separator size and design were selected, and a large effort was devoted to working with the NASA engineering team and its contractor, ZIN Technologies, to develop material and prepare for the Requirements Definition Review (RDR) and then the Science Review Document (SRD) and test matrix. This involved finalizing the requirements for all components of the system and defining the detailed conditions of the planned experiments in order to insure successful flight tests of the DYNASWIRL® phase separator onboard the ISS. Unfortunately, NASA decided later to not pursue the TPFSE experiments onboard the International Space Station (ISS). As a result, our interaction with ZIN Technology and the NASA engineers, who were working on preparing for the tests, ceased and our efforts in the last years reverted to more fundamental two-phase flow investigations of the separator aimed at expanding its field of applications. This meant that we were no longer tied to one TPFSE-compliant DYNASWIRL® configuration and this opened up our work to considering wider operating envelopes. Following this reorientation, we interacted with NASA personnel to seek other applications for the technology and reverted back to our science objectives of studying the details of the swirl separator technology. We adjusted the focus of the project to concentrate on the investigation of scale effects, effect of the various geometrical dimensions of the separator on the performance, and on the flow patterns in the DYNASWIRL® phase separator. This also enabled us to study not only the steady state behavior of the separator but also its transient behavior during start up procedures where very high void fractions (up to 100%) are involved. This was encouraged by the interest of the Space Technology Mission Directorate at NASA in the operation of the separator for cryogenic liquid/vapor separation, for instance during transfer from tank to tank. These developments resulted in a parallel NASA Phase I SBIR (Small Business Innovation Researc) in which we designed and built a stainless steel separator for high flow rates (max flow rate four times higher than the max flow rate planned for the TPFSE, 20 gpm vs. 5 gpm) and very low temperatures, which we tested in liquid nitrogen LN2. Extensive parallel water tests were also conducted, which fed into the present project and helped us make significant progress. These efforts were successful and we were just awarded a 2-year Phase II SBIR to further develop the separator for the cryogenic application. In the R&D work conducted in the last years of this project, the DYNASWIRL® operation limits, both in void fractions and flow rate, were very significantly extended. The dimensions of the separator, the exit orifice, and the slots were varied and their effects on the separation tested. Flow rates as high as 20 gpm (~76 lpm) were achieved with good separation in both water and liquid nitrogen. In both liquids, captured air or nitrogen vapors at the start of the transfer of the mixture between two tanks were removed successfully. In each case, the void fraction was temporarily close to

	100%. In addition to the above, significant effort was spent on studying scale effects on the operation of the separator. Three scales of the separator were built in Plexiglas and were extensively tested. The slots geometry, number, and flow passages areas, the liquid exit orifice, and height of the chamber was kept the same, while three inner chamber diameters were investigated: ~2.5 cm, ~5 cm, and ~10 cm. All operated properly, but there were differences in the flow field indicating scale effects. These were characterized using Particle Imaging Velocimetry (PIV).and pressure measurements as well as numerical simulations. The effects of the inserts upstream of the liquid exit orifice on the DYNASWIRL® operation and on the pressure drop in the system were also extensively studied. These inserts include the following: a) A vortex “Stopper,” which is a disk of diameter Dd placed at a distance Hs from the exit orifice of diameter d0. This disk plays two roles: 1) it stops or delays any long gaseous core from reaching the liquid exit orifice, 2) it is needed to structurally support the tubular membrane discussed below as Item c). b) A vortex “Killer,” which is a thin cruciform structure located between the “Stopper” and the liquid exit orifice, which practically stops the liquid rotation downstream of the “Stopper” while not generating any flow constrictions on the exiting liquid. c) A microporous gas permeable membrane rolled around a perforated tube, which allows extraction of the gas collected on the vortex axis in a gaseous core. The experimental and numerical investigations allowed us to characterize the effects of Dd/d0 and Hs/d0 on the strength of the vortex generated for a given flow rate. We found that, even though the presence of the “Stopper” reduces the vortex strength near the axis and increases pressure losses across the separator, these effects are less prominent when Dd/d0<1 and when Hs/d0>1. The study of the “Killer” was the most rewarding as it showed that a properly designed and dimensioned killer used in combination with the “Stopper” can reduce significantly the pressure drop across the separator. This was shown both experimentally and numerically and will be very helpful for future work with the separators. Increasing the limit of operation of the DYNASWIRL® benefitted also significantly for the presence of the “Killer” which reduced the pressure losses.
Bibliography Type:	Description: (Last Updated: 09/04/2025)
Abstracts for Journals and Proceedings	Loraine G, Hsiao C-T, Ma J, Chahine GL. "Study of the Geometric Scaling Effects on the DYNASWIRL® Phase Separator." 35th Annual Meeting of the American Society for Gravitational and Space Research, Denver, CO, November 20-23, 2019. Abstracts. 35th Annual Meeting of the American Society for Gravitational and Space Research, Denver, CO, November 20-23, 2019. , Nov-2019
Articles in Peer-reviewed Journals	Ma J, Hsiao CT, Chahine GL. "A physics based multiscale modeling of cavitating flows." Comput Fluids. 2017 Mar 2;145:68-84. https://doi.org/10.1016/j.compfluid.2016.12.010 ; PMID: 29720773; PMCID: PMC5927377. , Mar-2017